Seungyeon Lee

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RESEARCH INTERESTS

- **Spatiotemporal Network Analysis:** Time-series mobility modeling, cascading disruption tracking, and traffic flow dynamics under dynamic environments.
- **Dynamic Mobility Equity & Spatial Accessibility:** Time-varying accessibility modeling and hour-by-hour vulnerability assessment for transport-disadvantaged populations.
- **GeoAI & Predictive Simulation:** Graph neural networks (GNNs), spatiotemporal deep learning, and predictive traffic modeling for urban resilience.

EDUCATION

Kyung Hee University, Seoul, Republic of Korea **Mar 2023 – Feb 2025**
M.S. in Geography (Specialization: GIS) GPA: 4.25 / 4.3 (3.95 / 4.0)

- Thesis: *Spatiotemporal Network-Based Traffic Congestion Prediction Using Grid Data – Case Study of Atlanta, Georgia, USA*
- Advisor: Dr. Chul Sue Hwang

Kyung Hee University, Seoul, Republic of Korea **Mar 2019 – Feb 2023**
B.S. in Geography GPA: 3.87 / 4.3 (3.60 / 4.0)

- Teacher Certificate of Geography (Ministry of Education, Republic of Korea)

PUBLICATIONS

[5] Cho, Y., Seong, J. C., **Lee, S.**, & Hwang, C. S. (2025). A Study on the Correlation Between Intersection Traffic Environment Complexity and Traffic Accidents: A Case Study of Atlanta, Georgia, USA. *Journal of the Korean Geographical Society*, 60(4), 451–463. DOI: 10.22776/kgs.2025.60.4.451

[4] Seong, J. C., **Lee, S.**, Cho, Y., & Hwang, C. S. (2025). Beyond the Road: A Regional Perspective on Traffic Congestion in Metro Atlanta. *ISPRS International Journal of Geo-Information*, 14(2), 61. DOI: 10.3390/ijgi14020061

[3] **Lee, S.**, Hwang, T., Lee, W. D., & Hwang, C. S. (2024). Unveiling Spatial Disparities between Public Rental Housing and Young People's Daily Activity Spaces. *Journal of the Korean Geographical Society*, 59(4), 573–584. DOI: 10.22776/kgs.2024.59.4.573

[2] Kim, S., **Lee, S.**, Kim, M., Lee, C., & Kang, J. Y. (2024). Vulnerability Assessment for Pediatric Healthcare Services Using Agent-based Modeling: A Case Study of Four Districts in Southwest Seoul. *Journal of the Korean Geographical Society*, 59(2), 196–209. DOI: 10.22776/kgs.2024.59.2.196

[1] **Lee, S.**, Hwang, T., Lee, W. D., & Hwang, C. S. (2023). Measuring Spatial Associations of Intercity Flows between Depopulation Regions Considering Nearest Neighborhoods. *Journal of the Korean Geographical Society*, 58(6), 644–656. DOI: 10.22776/kgs.2023.58.6.644

RESEARCH EXPERIENCE

Research Assistant **Feb 2022 – Present**
Kyung Hee University, Republic of Korea (Supervisor: Dr. Chul Sue Hwang)

- Researched spatiotemporal network modeling, GeoAI frameworks, dynamic mobility analysis, and spatial accessibility.
- Developed automated spatial data processing and predictive modeling pipelines using Python (GeoPandas, PyTorch), QGIS, and ArcGIS Pro.
- Participated in the project *Development of Support Strategies for Depopulated Regions in Republic of Korea* funded by KATRI (Jun 2022 – Sep 2022).
- Co-authored 5 peer-reviewed journal publications in spatiotemporal network analytics and urban spatial equity.

Visiting Researcher **Jun 2024 – Feb 2025**
University of West Georgia, USA (Supervisors: Dr. Chul Sue Hwang and Dr. Jeong Chang Seong)

- Participated in the *Global ICT Professional Training Program* funded by IITP, Republic of Korea.
- Conducted research on detecting urban traffic risk areas using spatial big data and GeoAI frameworks.
- Published a co-authored research paper in the *ISPRS International Journal of Geo-Information (IJGI).*

CONFERENCE PRESENTATIONS

[10] **Lee, S.**, Hwang, C. S., & Seong, J. C. (2025, June). *Traffic congestion prediction using Spatio-Temporal Network-based GeoAI framework: A case study of Atlanta* [Paper presentation]. Proceedings of the Korean Geographical Society Conference, Republic of Korea.

[9] Cho, S., Hwang, T., Lee, Y., **Lee, S.**, & Hwang, C. S. (2024, April). *Nonlinear analysis of high-resolution urban thermal environment using random forest method: Seoul, Korea* [Poster presentation]. 2024 AAG Annual Meeting, Honolulu, HI, United States.

[8] **Lee, S.**, Hwang, T., Lee, W. D., & Hwang, C. S. (2024, April). *Unveiling Spatial Mismatch between Public Rental Housing and Youth's Activity* [Paper presentation]. 2024 AAG Annual Meeting, Honolulu, HI, United States.

[7] **Lee, S.**, Hwang, T., Lee, W. D., & Hwang, C. S. (2023, December). *Measuring Spatial Associations of Intercity Flows between Depopulation Regions Considering Nearest Neighborhoods* [Paper presentation]. Proceedings of the Korean Geographical Society Conference, Republic of Korea.

[6] Hwang, T., Kim, K., Lee, Y., Cho, S., **Lee, S.**, Hong, H., & Hwang, C. S. (2023, June). *Exploring spatial patterns of bus and shared bike OD using BiFlowLISA* [Paper presentation]. Proceedings of the Korean Geographical Society Conference, Republic of Korea.

[5] **Lee, S.**, Hwang, T., Hwang, C. S., & Lee, W. (2023, June). *Exploring similarities and heterogeneities of spatial flows between regions using spatial analysis techniques* [Paper presentation]. Proceedings of the Korean Geographical Society Conference, Republic of Korea.

[4] Yoo, S., **Lee, S.**, & Lee, W. (2023, June). *Current status of population change by depopulated areas and counter-measures* [Paper presentation]. Proceedings of the Korean Geographical Society Conference, Republic of Korea.

[3] Ji, S., **Lee, S.**, Hwang, T., & Lee, W. (2022, September). *Development of an open-source package for measuring transport accessibility using standard node-links* [Paper presentation]. Proceedings of the Korean Society of Transportation Conference, Republic of Korea.

[2] **Lee, S.** (2022, June). *Analysis of fire-vulnerable areas in Dongdaemun-gu: Service area analysis for selecting fire-vulnerable areas in Dongdaemun-gu* [Poster presentation]. Proceedings of the Korean Geographical Society Conference, Republic of Korea.

[1] Kim, S., Min, Y., Park, S., **Lee, S.**, & Jeong, S. (2021, June). *Creating a 10-minute 'Sulse-gwon' in our neighborhood: Analysis of living SOC status and resident perception in Dongdaemun-gu* [Poster presentation]. Proceedings of the Korean Geographical Society Conference, Republic of Korea.

GRANTS, SCHOLARSHIPS & AWARDS

Academic Excellence Scholarship for Top Graduates **Mar 2023 – Feb 2025**
Kyung Hee University, South Korea **Grant: ∼$8,887 USD**

- Awarded full tuition coverage based on outstanding cumulative GPA (3.8+/4.3). Total: 12.8M KRW.

Global Major Field Training Grant **May 2019**
Kyung Hee University, South Korea **Grant: ∼$2,073 USD**

- Overseas academic training program at the University of West Georgia, USA. Total: 3M KRW.

2nd Place, Spatial Analysis Research Contest **Dec 2022**
Project: Analysis of Fire Vulnerable Areas in Dongdaemun-gu, Seoul

1st Place, Spatiotemporal Information Application Paper Contest **Feb 2022**
Project: Analysis of Hourly Fire Accessibility Through the Construction of Road Networks

3rd Place, Spatial Big Data Academic Festival & Competition **Dec 2019**
Project: Analysis of Spatial Characteristics of Domestic City Brand Slogans

SKILLS & METHODOLOGIES

- **Languages:** English (IELTS Overall 7.5), Korean (Native)
- **Spatial Statistics & Analysis:** Spatial Autocorrelation (Moran's I, LISA), Spatial Regression (GWR, MGWR), Spatial Clustering (Gi^*), Spatio-Temporal Cube, Road Network Analysis (Dijkstra's), Accessibility Modeling (2SFCA, E2SFCA), Agent-Based Modeling
- **GeoAI & Machine Learning:** Graph Neural Networks (GCN, TGCN, STGCN, GC-LSTM), LSTM, Random Forest, XGBoost
- **Software & Programming:** Python, R, PyTorch, AnyLogic, ArcGIS Pro, QGIS, Pix4D Mapper, ERDAS Imagine

TEACHING & ACADEMIC SERVICE

Teaching Assistant, UWG Geospatial Workshop **Jul 2024**
University of West Georgia, USA (Supervisors: Dr. Chul Sue Hwang and Dr. Jeong Chang Seong)

- Mentored undergraduate student teams and provided hands-on guidance on ArcGIS Pro and geospatial data analysis.

Teaching Assistant, Applications of Geographic Information and Technology **2022 – 2023**
Kyung Hee University, Republic of Korea

- Provided technical tutoring to undergraduate students on drone imagery processing and GIS tools (Pix4D, ArcGIS Pro).

Academic Program Assistant, Geospatial Information Innovation Cultivation Program **Jul 2022 – Feb 2025**
Kyung Hee University, Republic of Korea (Funded by Korea Land and Geospatial InformatiX Corporation)

- Coordinated academic activities and mentored undergraduate students in practical spatial data analysis.

Conference Volunteer, The Korean Geographical Society & CJK Conference **2022 – 2023**

- Supported session management for the 2022 KGS Annual Conference and 2023 CJK Regional Conference.

Academic Club Member, Human Geography Club **2019 – 2021**

- Conducted collaborative spatial data projects and human geography theory seminars.

REFERENCES

- **Dr. Chul Sue Hwang** – Professor, Department of Geography, Kyung Hee University
  Relationship: M.S. Thesis Advisor / B.S. Academic Advisor — *Email:* hcs@khu.ac.kr
- **Dr. Jeon-Young Kang** – Associate Professor, Department of Geography, Kyung Hee University
  Relationship: M.S. Thesis Committee — *Email:* geokang@khu.ac.kr
- **Dr. Jeong Chang Seong** – Professor, School of Field Investigations & Experimental Sciences, University of West Georgia
  Relationship: Research Supervisor (J-1 Exchange Program) — *Email:* jseong@westga.edu